Values of *LET* in water are given for various particles and energies in Table 7.2. For the same energy and the same absorbing material, the *LET* values increase in the order:

high energy electrons (also approximatively γ-rays
< β-particles (also approximately soft X-rays)
< protons
< deuterons
< heavy ions (ions of N, O, etc.)
< fission fragments

The relationship between *LET*, which refers to the energy absorbed in matter, and the stopping power, which refers to the energy loss of the particle, is

$$dE_{loss}/dx = dE_{abs}/dx + E_x \tag{8.6}$$

The difference E_x in these two energy terms is related to the energy loss by electromagnetic radiation (mainly bremsstrahlung).

8.1.2. Uncharged radiation

When neutrons or photons having the incident particle energy E_{in} are absorbed, a certain fraction of energy E_{tr} is transferred into kinetic energy of charged particles when traversing the distance dx. We can define an *energy transfer coefficient* as

$$\mu_{tr} = E_{in}^{-1} dE_{tr}/dx \left(m^{-1}\right) \tag{8.7}$$

If we neglect the bremsstrahlung associated with the absorption of the secondary charged particles formed in the initial absorption processes, E_{tr} is the energy absorbed (designated as E_{abs}), and we can write Eqn (8.7) as

$$\mu_a = E_{in}^{-1} dE_{abs}/dx \left(m^{-1}\right) \tag{8.8}$$

8.2. RADIATION TRACKS

The energy lost when a high energy charged particle is slowed in matter gives rise to a trail (or, more commonly, track) of ionized and excited molecules along the path of the particle; Figure 8.1 schematically depicts a track. Cloud chamber photographs of such tracks are shown in Figure 7.5. A photon imparts a large fraction of its energy to a single electron which subsequently ionizes and excites many other molecules along its path. The absorption of any type of ionizing radiation by matter thus leads to the formation of tracks of ionized and excited species. Whereas these species generally are the same in a particular sample of matter regardless of the type of radiation, the tracks may be sparsely or densely populated by these species. Expressions such as linear energy transfer (*LET*) and stopping power ($\hat{S}$) are based on the implicit assumption of a continuous slowing down process and thus gives quantitative information on the average energy loss but only qualitative information on the densities of reactive species.

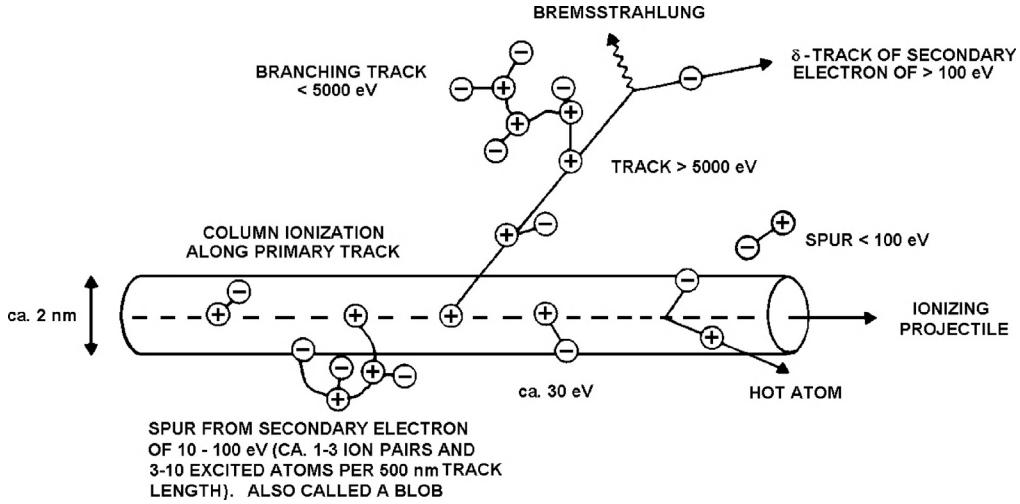


Figure 8.1 Track formed by energetic ionizing particle in condensed matter. Distances between ion pairs along track are: ~1000 nm γ, 500-100 nm fast electron, ~1 nm slow electron and α.

Most of the radiation-matter interactions are involving the electrons rather then the nucleus itself. As seen previously these interactions with the electrons may occur in several ways. Thus, electrons liberated in the ionization process will have greatly varying kinetic energy. If the energy of these secondary electrons is relatively low (<100 eV) their range in liquids and solids is short and the ionizations and excitations caused by these electrons take place close to the primary ionizations leading to the formation of small *spurs* containing ionized and excited species. Secondary electrons with high kinetic energy form tracks of their own branching from the primary track. Such electrons are called *δ-rays*. For high energy electrons, spurs are formed at well separated intervals along the track whereas for densely ionizing radiation such as α particles, protons and recoil atomic ions the spurs overlap and form *columns* of excited and ionized species. Differences in chemical and biological effects caused by different radiations reflects the varying track structures.

The tracks of X-ray and γ-radiation result in tracks of fast electrons. The energy of these fast electron is consumed by the formation of spurs (6−100 eV), blobs (100−500 eV) and short tracks (500−5000 eV), see Figure 8.1. For a primary electron in the range 10^4−10^7 eV the distribution of energy of secondary electrons is approximately 40% <3.4 eV, 20% 3.4−6.8 eV, 18% 6.8−13.5 eV and 12% 13.5−27.1 eV. The radius of the tracks core at low electron velocities is of the order of 1 nm. Thus it is possible to calculate in e.g. a solution how many molecules will be affected by the tracks based in the concentration on a certain molecule.

The energy transferred to the electrons by an energetic ion depends on the mass, the charge (Z) and velocity (v) of the ion, cf. Eqn (7.14): the probability of interaction is proportional to the ratio Z^2/mv^2. Heavy ions produce track structures similar to those of fast

electrons. However, the spurs are quite close to each other. For the case of α-particles, the original distance between spurs is of the order of fractions of a nanometer. Immediately at their formation they comprise a continuous cylindrical column. The column consists of a dense core surrounded by a more diffuse shell of tracks of high energy δ-electrons.

8.3. RADIATION DOSE AND RADIATION YIELD

The oldest radiation unit still in use is the *roentgen* (R). It applies only to photons and is defined as the *exposure*. The exposure is the energy flux of the unperturbed photon radiation when hitting matter. 1 R is the exposure that in air produces ion pairs with total charge per unit mass of 2.58×10^{-4} C/kg. This corresponds to 1.61×10^{15} ion pairs per kg air or 8.8×10^{-3} J/kg absorbed energy at an average ion pair formation energy of 34 eV. The roentgen has been in use for more than 60 years and is still not uncommon in the medical profession in some countries.

The *absorbed dose* (D) is the amount of radiation energy absorbed per unit mass:

$$D = \mathrm{d}E_{\mathrm{abs}}/\mathrm{d}m \tag{8.9}$$

According to Eqn (8.1), in the absense of nuclear transformations, $E_{\mathrm{abs}} = E_{\mathrm{in}} - E_{\mathrm{out}}$. The SI unit is the *Gray* (Gy)

$$1\ \mathrm{Gy} = 1\ \mathrm{J/kg} \tag{8.10}$$

An old, more or less obsolete, unit is the *rad* (for radiation absorbed dose)

$$1\ \mathrm{Gy} = 100\ \mathrm{rad} \tag{8.11}$$

The rad was proposed in 1918 as "that quantity of X rays which when absorbed will cause the destruction of the malignant mammalian cells in question..." Formally it was defined in 1953 as the dose causing 100 ergs of energy to be absorbed by one gram of matter. An erg is the unit of energy and mechanical work in the old centimetre-gram-second (CGS) system of units. The name comes from the greek word for work, ergon. One erg is the amount of work done by a force of one dyne exerted for a distance of one centimeter. The dyne also originates from a greek word, namely dynalis meaning force or power. The dyne is defined as "the force required to accelerate a mass of one gram at a rate of one centimeter per second squared" Although these units may still be found in the literature their use is generally discouraged in education for other pursposes than for relating old measures to units in the SI system. In some countries, however, they are still used as an industrial standard.

The *dose rate* is the absorbed dose per unit time. The SI unit is Gray per second (Gy/s).

The *specific γ-ray dose rate,* $\dot{D}$, is a practical measure for estimation of the radiation hazard to people from γ-emitting radionuclides:

$$\dot{D} = A\, r^{-2} \Sigma\, n_i\, k_i\, B_i\, \mathrm{e}^{-\mu_i x} \tag{8.12}$$